BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q.PAPER – II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8477

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The rest mass energy of electron positron pair is :

(A) 0.51 MeV
(B) 0.71 MeV
(C) 1.02 MeV
(D) 2 MeV

Q2. The SI unit of impedance is :

(A) Ohm
(B) Farad
(C) Volt
(D) Ampere

Q3. To convert galvanometer into voltmeter, high resistance is connected to the galvanometer in :

(A) Parallel
(B) Series
(C) Anti parallel
(D) Perpendicular

Q4. In transistor, concentration of impurity is highest in :

(A) Collector
(B) Emitter
(C) Base and collector
(D) Base

Q5. At high frequency, RLC series circuit behaves like :

(A) R-C circuit
(B) R-L circuit
(C) RLC series circuit
(D) L-C circuit

Q6. If electric and gravitational forces on an electron balance each other, then electric field intensity will be :

(A) mgq
(B) q/mg
(C) mg/q
(D) $q / 4\pi\varepsilon_0 r^2$

Q7. The temperature of steam coming out of turbine in nuclear reactor is :

(A) 200 °C
(B) [illegible in source scan]00 °C
(C) 600 °C
(D) 1300 °C

Q8. The dimensions of motional emf are same as that of :

(A) Magnetic induction
(B) Magnetic flux
(C) Potential difference
(D) Magnetic force

Q9. The value of Stefen's constant 'σ' is given by :

(A) $5.67\times10^{-8}\ Wm^{-2}K^{-2}$
(B) $5.67\times10^{-8}\ Wm^{-2}K^{-4}$
(C) $5.67\times10^{-8}\ Wm^{2}K^{2}$
(D) $5.67\times10^{-8}\ W^{2}m^{2}K^{-2}$

Q10. A charge of 4C is placed in the field of intensity $8NC^{-1}$. The force on the charge is :

(A) 2 N
(B) 4 N
(C) 16 N
(D) 32 N

Q11. The example of crystalline solid is :

(A) Zirconia
(B) Natural rubber
(C) Polystrene
(D) Nylon

Q12. Heat sensitive resistors are called :

(A) Resistor
(B) Thermistor
(C) Inductor
(D) Capacitor

Q13. The atoms can reside in metastable state for about :

(A) $10^{-2}s$
(B) $10^{-3}s$
(C) $10^{-4}s$
(D) $10^{-8}s$

Q14. X = A + B is the mathematical notation for :

(A) OR gate
(B) NOR gate
(C) NOT gate
(D) NAND gate

Q15. Binding energy per nucleon for isotope iron-58 has a value of :

(A) 6.6 MeV
(B) 7.7 MeV
(C) 8.8 MeV
(D) 9.9 MeV

Q16. For step up transformer :

(A) Ns < Np
(B) Ns > Np
(C) Ns = Np
(D) Ns ≥ Np

Q17. Brightness of screen of CRO is controlled by :

(A) Grid
(B) Anode
(C) Cathode
(D) Filament

Essay Type (Group I Subjective)

Time Allowed: 2.40 hours | **Maximum Marks:** 68

Note: Group I Subjective (Essay Type) 2024 is missing from the source: the scan served on the source page is the Inter Part-I (11th class) Physics paper mislabeled as Inter Part-II. The correct 12th-class Group I Subjective scan is not available on ilmkidunya/tayarri.

GROUP – II

Q.PAPER – II (Objective Type)

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8476

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. What is the critical temperature of Yttrium Barium Copper Oxide ($YBa_2Cu_3O_7$) :

(A) 4.2 K
(B) 110 K
(C) 163 K
(D) 7.2 K

Q2. One henry (H) is defined as :

(A) $1H = 1VS^{-1}A^{-1}$
(B) $1H = 1VSA$
(C) $1H = 1VSA^{-1}$
(D) $1H = 1VS^{-1}A$

Q3. Choose the photon of highest energy among the following :

(A) X-rays
(B) Infrared
(C) Radiowaves
(D) Gamma rays

Q4. A particle having a charge of 2e falls through a potential difference of 3V. The energy acquired by it will be :

(A) 5 eV
(B) 1.5 eV
(C) 6 eV
(D) 0.6 eV

Q5. SI unit of equivalent dose is :

(A) Sievert
(B) Gray
(C) Rad
(D) Curie

Q6. If peak value of AC voltage is 100 V, then the peak to peak value will be :

(A) 200 V
(B) 50 V
(C) 70 V
(D) 1000 V

Q7. The direction of magnetic lines of force around a straight current carrying conductor is found by :

(A) Ampere's law
(B) Coulomb's law
(C) Lenz's law
(D) Right hand rule

Q8. Which of the following is the correct relation between electric intensity E and potential difference ΔV :

(A) $E = -\Delta V/\Delta r$
(B) $\Delta V = E/\Delta r$
(C) $E = \Delta V + \Delta r$
(D) $E = \Delta V^2/\Delta r^2$

Q9. Which of the following requires no external bias for its operation :

(A) LED
(B) Photo diode
(C) Photo-voltaic cell
(D) Transistor

Q10. The energy of Kα X-rays is :

(A) $hf_{k\alpha} = E_M - E_K$
(B) $hf_{k\alpha} = E_L - E_K$
(C) $hf_{k\alpha} = E_K - E_M$
(D) $hf_{k\alpha} = E_N - E_M$

Q11. The power factor of a series resonance circuit at resonance frequency is :

(A) Zero
(B) Infinite
(C) 2
(D) 1

Q12. In AVO meter, the part which connects the galvanometer with the relevant measuring circuit is known as :

(A) Range switch
(B) Diode
(C) Ground
(D) Function selector

Q13. How much time is required for the complete decay of a radioactive element :

(A) Five half lives
(B) Two half lives
(C) Ten half lives
(D) Infinite

Q14. Choose the device which converts electrical energy into mechanical energy :

(A) Motor
(B) Generator
(C) Transformer
(D) Inductor

Q15. The current-voltage graph of an ohmic material is :

(A) Curve
(B) Straight line
(C) Parabolic
(D) Circular

Q16. The phase shift between the input and output of a common-emitter transistor amplifier is :

(A) 90°
(B) 180°
(C) 60°
(D) 45°

Q17. Which of the following factor is called Compton Wavelength :

(A) $h / m_o c$
(B) $m_o c / h$
(C) hc / m_o
(D) m_o / hc

PAPER – II (Essay Type)

Time Allowed: 2.40 hours | **Maximum Marks:** 68

Q2. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) Do electrons tend to go to region of high potential or of low potential?

(ii) How can you identify that which plate of a capacitor is positively charged?

(iii) Define electric potential. Write its SI unit.

(iv) How Millikan concluded that minimum value of the charge is the charge on an electron?

(v) Why a voltmeter should have a very high resistance?

(vi) Why does the picture on a TV screen become distorted when a magnet is brought near the screen?

(vii) State Ampere's law. Write its mathematical form.

(viii) How the path of electrons is made visible in glass tube to measure e/m ratio?

(ix) What do we mean by the term critical mass?

(x) How can radioactivity help in treatment of cancer?

(xi) How do gamma rays photon interact with matter at low and high energy?

(xii) How did James Chadwick discover a neutron?

Q3. (Marks 16) Write short answers to any EIGHT (8) questions :

(i) How can a rheostat be used as a potential divider? Draw also diagram.

(ii) Do bends in a wire affect its electrical resistance? Explain.

(iii) Explain thermistors, their construction and shapes.

(iv) Define inductive reactance and capacitive reactance. Also write mathematical formula of each.

(v) At what frequency will an inductor of 1 H have a reactance of 500 Ω ?

(vi) How reception of a particular radio station is selected on your radio set?

(vii) Give a comparison of crystalline and amorphous solids briefly.

(viii) Differentiate between elasticity and plasticity.

(ix) What is meant by paramagnetic and ferromagnetic substances?

(x) What is the effect of forward biasing and reverse biasing of a diode on the width of depletion region?

(xi) Draw circuit diagram of full wave rectifier.

(xii) Why is the base current in a transistor very small?

Q4. (Marks 12) Write short answers to any SIX (6) questions :

(i) State the Lenz's law and explain the significance of -ve sign in Faraday's law.

(ii) Does the induced emf always acts to decrease the magnetic flux through a circuit?

(iii) What is the efficiency of a transformer? Describe methods to increase it.

(iv) As a solid is heated and begins to glow, why does it first appear red?

(v) Write two properties of intensity distribution diagram.

(vi) When does the light behave as a particle and when does it behave as a wave?

(vii) Which photon, red, green or blue carries the most (a) energy (b) momentum.

(viii) Bohr's theory of hydrogen atom is based upon several assumptions. Do any of these contradict classical physics?

(ix) Differentiate between spontaneous and stimulated emissions.

> Note: SECTION – II

> Note: Note : Attempt any THREE questions.

Q5.

(a) [5] Derive an expression for the energy stored in a capacitor.

(b) [3] The resistance of an iron wire at 0 °C is 1×10^{4} Ω. What is the resistance at 500 °C if the temperature co-efficient of resistance of iron is 5.2×10^{-3} K^{-1}?

Q6.

(a) [5] State Ampere's law. Calculate the magnetic field due to current carrying solenoid.

(b) [3] A circular coil has 15 turns of radius 2 cm each. The plane of the coil lies at 40° to the uniform magnetic field of 0.2 T. If the field is increased by 0.5 T in 0.2 s, find the magnitude of induced emf.

Q7.

(a) [5] Discuss the behaviour of an inductor in an A.C. circuit and write expression for inductive reactance.

(b) [3] In a certain circuit, the transistor has a collector current of 10 mA and a base current of 40μA. What is the current gain of transistor?

Q8.

(a) [5] What is meant by strain energy? Derive the relation for strain energy in deformed materials.

(b) [3] X-rays of wavelength 22 pm are scattered from a carbon target. The scattered radiation being viewed at 85° to the incident beam. What is Compton Shift?

Q9.

(a) [5] How de-Broglie's interpret Bohr's 2nd postulate that an angular momentum is equal to integral multiple of h/2π?

(b) [3] A sheet of lead 5.0 mm thick reduces the intensity of a beam of ɣ-rays by a factor 0.4. Find half value thickness of lead sheet which will reduce the intensity of half of its initial value.